

SECTION B

9 (long)

(a) The complex variable z satisfies the equation

$$|z - 4| = \lambda |z - z_0|$$

where $z_0 = 2 + 2i$.

(i) Find the equation of the locus of z for $\lambda = 1$ and describe its geometry. [3]

(ii) Find the equation of the locus of z for $\lambda = \sqrt{2}$ and describe its geometry. [4]

(b) Integrate the expression

$$\int e^{ax} \cos(bx) dx$$

by expressing the integrand as the real part of $e^{ax} e^{ibx}$. [6]

(c) (i) Find the roots of $z^{1/3}$ where

$$z = 2 + 2i$$

and plot all of these roots in the complex plane on an Argand diagram. [8]

(ii) Solve the equation

$$z^6 - 4z^3 + 8 = 0$$

and plot the position of all its roots in the complex plane on an Argand diagram. [9]